INCH-POUND

MIL-PRF-9439F
27April 2009
SUPERSEDING
MIL-PRF-9439E
17 July 1998

PERFORMANCE SPECIFICATION

VALVES, OXYGEN CYLINDER, HIGH PRESSURE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

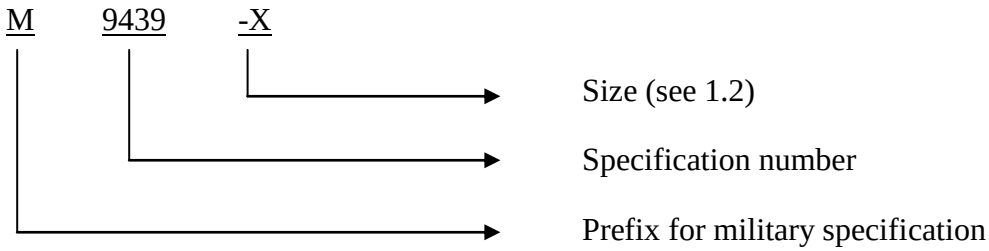
1. SCOPE

1.1 Scope. This specification covers manually operated, high pressure, aeronautical, oxygen cylinder valves.

1.2 Classification. Valves covered by this specification are classified according to the following sizes, as defined in figure 1:

Size 1	0.5 inch aeronautical pipe thread (ANPT) – inlet
Size 2	1.0 inch ANPT – inlet

1.3 Part or Identifying Number (PIN). The PIN to be used for valves acquired to this specification is created as follows:



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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following standard forms a part of this document to the extent specified herein. Unless otherwise specified, the issue of this document are those cited in the solicitation or contract.

FEDERAL STANDARD

FED-STD-H28/9	- Screw-Thread Standards for Federal Services Section 9 Gas Cylinder Valve Outlet and Inlet Threads

(Copies of this document are available online at http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B117	- Salt Spray (Fog) Apparatus, Operating

(ASTM documents may be obtained at http://www.astm.org/ or addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.4	- Sampling Procedures and Tables for Inspection by Attributes

(ASQ documents may be obtained at http://www.asq.org/ or from American Society for Quality, P.O. Box 3005, Milwaukee, WI 53201-3005 or 600 North Plankinton Avenue, Milwaukee, WI 53203.)

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SAE INTERNATIONAL

SAE AS8010	- Aviator's Breathing Oxygen Purity Standard
SAE AS71051	- Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT-Design and Inspection Standard

(SAE documents may be obtained at http://www.sae.org/ or from Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Materials. All materials shall be corrosion-resistant or suitably treated to resist corrosion due to electrolytic decomposition, salt spray, and any other atmospheric condition that may be encountered during operational use or storage. The use of toxic chemicals, hazardous substances, or ozone depleting chemicals shall be avoided, whenever feasible.

3.3.1 Safety specific materials. All materials used shall be capable of operating safely in a pure oxygen environment without creating a fire or health hazard. Lubrication shall be compatible with pure oxygen environment (see 6.5). Aluminum alloys shall not be used.

3.4 Interface.

3.4.1 Dimensions. The dimensions of the valve shall be in accordance with figure 1.

3.4.2 Valve outlet. The valve outlet shall be in accordance with FED-STD-H28/9 (see figure 1).

3.4.3 Threads. Inlet pipe threads shall be in accordance with SAE-AS71051.

3.4.4 Handwheel markings. Handwheel shall be marked as shown in figure 1.

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3.5 Performance.

3.5.1 General. The valve shall be manually operated, positive shut-off type that controls oxygen flow to and from 2,100 psig oxygen cylinders. The valve shall be fully opened by turning the handwheel ¼ to one turn counterclockwise (see figure 1).

3.5.2 Weight. The weight of the valve, excluding the cap, shall be in accordance with figure 1.

3.5.3 Opening and closing torque. The torque required to open and close the valve shall be not greater than 60 inch-pounds when the valve is pressurized to 2,200 psig.

3.5.4 Flow. The valve flow shall be not less than 708 standard liters per minute (SLPM) (see 6.4).

3.5.5 Leakage. The valve outlet shall not leak when the valve is in the closed position and the valve stem shall not leak in any handwheel position.

3.5.6 Proof pressure. The valve shall not leak or distort at a proof pressure of 3,000 psig.

3.5.7 Endurance. The valve shall not leak or distort during 250 cycles of opening and closing with an inlet pressure of 2,200 psi.

3.5.8 Cleanliness. All surfaces shall be free of visible particulates (50 microns and larger), free of visible fluorinated lubricants, and free of hydrocarbon contamination to a level not greater than 3 milligrams per square foot (mg/ft^2) (see 6.3).

3.5.9 Lubrication. Internal, lubricated parts of the valve shall not contact the oxygen flow (see 6.5).

3.5.10 Contamination prevention device. The valve shall be provided with a device, such as a siphon tube, to diffuse suction within the oxygen cylinder and prevent foreign material from being carried into the valve (see 6.7).

3.5.11 Overpressure safety device. The valve shall be provided with a device, such as a frangible safety disc, to ensure rapid dumping of pressure should the internal pressure of the valve rise above 2,700 psig. The device shall activate between 2,700 and 3,000 psig and shall not leak below 2,700 psig.

3.5.12 Oxygen bomb. There shall be no evidence of charring or deterioration of the valve components when subjected to a pure oxygen surge pressure of 2,300 psig for 30 seconds.

3.5.13 Cap. A cap shall be provided for the outlet of the valve. The cap shall not leak when installed on the valve outlet and 2,200 psig is applied internally. The cap shall be tethered to the valve and the tether shall withstand 100 lbs. of pull.

3.6 Environmental conditions. The valve shall operate in the following conditions:

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a. Temperature: -65° to 160°F.
b. Vibration: Frequencies from 10 to 40 Hz at a double amplitude of 0.018 to 0.020 inch.

3.7 Item identification. Item shall be marked for identification with the following information:

a. Item name.
b. Manufacturer's name.
c. Manufacturer's CAGE code.
d. Manufacturer's part number (P/N).
e. MIL-PRF-9439.

3.8 Interchangeability. All parts having the same number manufacturer's part number shall be functionally and dimensionally interchangeable.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspections. First article inspection shall be performed on four valves of each size, and shall include all the tests described (see 4.6).

4.3 Conformance inspection. Conformance inspection shall include the tests in 4.3.1 and 4.3.2.

4.3.1 Individual tests. Each valve shall be subjected to the following tests:

a. Examination (see 4.6.1).
b. Opening and closing torque (see 4.6.4).
c. Leakage (see 4.6.6).

4.3.2 Sample tests. After completion of the individual tests (see 4.3.1), a sample of valves shall be selected in accordance with ASQ-Z1.4 at a normal inspection level unless modified in the procurement document (see 6.2), and be subjected to the following examination and tests:

a. Dimensions (see 4.6.2).
b. Flow (see 4.6.5).
c. Proof pressure (see 4.6.9).

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4.4 Test conditions.

4.4.1 Test medium. Unless otherwise specified in the test description, oxygen conforming to SAE-AS8010, Type I, shall be used.

4.4.2 Temperature and pressure. Unless otherwise specified in the test description, tests shall be conducted at ambient temperature and pressure.

4.5 Requirements cross-reference matrix. Table I provides a cross-reference matrix of the section 3 requirements tested or verified in the paragraphs below.

TABLE I. Requirements cross-reference matrix.

Requirement	Verification	Requirement	Verification
3.1	4.2	3.5.6	4.6.9
3.3	4.6.1, 4.6.13, 4.6.14	3.5.7	4.6.10
3.3.1	4.6.13	3.5.8	4.6.3
3.4.1	4.6.2	3.5.9	4.6.13
3.4.2	4.6.2	3.5.10	4.6.1
3.4.3	4.6.2	3.5.11	4.6.12
3.4.4	4.6.1	3.5.12	4.6.13
3.5.1	4.6.4, 4.6.5	3.5.13	4.6.1, 4.6.6, 4.6.15
3.5.2	4.6.1	3.6	4.6.7, 4.6.8, 4.6.11
3.5.3	4.6.4	3.7	4.6.1
3.5.4	4.6.5	3.8	4.6.1
3.5.5	4.6.6		

4.6 Tests.

4.6.1 Examination. The valve shall be inspected to determine conformance of materials, weight, cap, tether, contamination prevention device, markings, interchangeability, and item identification.

4.6.2 Dimensions. The valve shall be inspected to determine conformance to dimensions, threads, and outlet requirements.

4.6.3 Cleanliness. All surfaces shall be free of visible particulates (50 microns and larger), free of visible fluorinated lubricants, and free of hydrocarbon contamination to a level not greater than 3 mg/ft^2 (see 6.4). Cleanliness of the surfaces shall be demonstrated by industrially accepted methods and these cleaning and verification methods shall be identified (see 6.2 and 6.3).

4.6.4 Opening and closing torque. With the valve closed, the outlet uncapped, and a minimum pressure of 2,200 psig applied to the valve inlet, the valve shall be turned to the fully open position (¼ to one turn). With gas flowing through the valve, the valve shall then be turned

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to the closed position. The torque required at the handwheel to open or close the valve shall be not greater than 60 inch-pounds.

4.6.5 Flow. A constant gas pressure of 100 psig shall be applied to the valve inlet and a constant back pressure of 30 inches of mercury shall be applied at the valve outlet (see 6.4). The flow through the fully open valve outlet shall be not less than 708 SLPM. Flow through the valve shall also be reversed and verified.

4.6.6 Leakage. With the valve closed, the outlet capped, and a minimum pressure of 2,200 psig being maintained at the valve inlet, the valve shall be fully opened (¼ to one turn) and immersed in water. There shall be no leakage from the valve stem or outlet. While immersed, the valve shall be turned to the half-closed position. There shall be no leakage from the valve stem or outlet. The valve shall then be closed with a torque not greater than 60 inch-pounds, the outlet uncapped and the valve immersed in water. After immersion for 1 minute, there shall be no leakage from the outlet. Leakage is indicated by a breakaway of bubbles (see 6.9).

4.6.7 High temperature exposure. The valve shall be exposed to a temperature not less than 160°F for 3 hours. After this exposure and while maintained at this temperature, the valve shall be subjected to the torque, flow, and leakage tests (see 4.6.4, 4.6.5, and 4.6.6).

4.6.8 Low temperature exposure. The valve shall be exposed to a temperature not greater than -65°F for 3 hours. After this exposure and while maintained at this temperature, the valve shall be subjected to the torque, flow, and leakage tests (see 4.6.4, 4.6.5, and 4.6.6). The valve shall be immersed in a solution during the leakage test that will not freeze at -70°F and has a viscosity not greater than 1 centipoise at 20°C.

4.6.9 Proof pressure. The valve shall be closed with a torque not greater than 60 inch-pounds, the outlet uncapped, and the overpressure safety device replaced by a solid plug or cap. A water pressure of 3,000 psig shall be applied to the valve inlet. While the pressure is being maintained for 5 minutes, the valve shall not leak or distort. Then, the valve outlet shall be capped and 3,000 psig water pressure applied to the valve inlet. While maintaining the pressure for 5 minutes, the valve shall be turned to the open position and examined for leakage or distortion. The valve shall not exhibit any evidence of leakage or distortion, or require a torque greater than 60 inch-pounds to open the valve.

4.6.10 Endurance. With the outlet of the valve capped, and a minimum gas pressure of 2,200 psig at the valve inlet, the valve shall be cycled 250 times at a constant rate of one cycle per minute. Turning the handwheel from the closed position to the opened position and back to the closed position shall constitute one cycle. After completing 250 cycles, the valve shall be subjected to the torque and leakage tests (see 4.6.4 and 4.6.6).

4.6.11 Vibration. The valve shall be vibrated along each of three mutually perpendicular axes of the valve for 1 hour with the frequencies being varied from 10 to 40 Hz, and back to 10 Hz in approximately 1 minute intervals at a double amplitude of 0.018 to 0.020 inch. After the third axis is cycled, the valve shall be closed, the outlet capped, and a gas pressure of 2,200 psig applied to the inlet. With the gas pressure maintained the valve shall be vibrated as specified

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above for 30 minutes on each axis. After approximately 4.5 hours of total vibration, the valve shall be subjected to the torque and leakage tests specified (see 4.6.4 and 4.6.6).

4.6.12 Overpressure safety device. While being exposed to a constant temperature not less than 160°F, the valve shall then be subjected to a minimum internal gas pressure of 2,200 psig for 30 seconds. The internal pressure shall then be raised in increments of approximately 50 psig and maintained at each pressure for 30 seconds. The pressure shall be raised in this manner until the overpressure safety device activates. The activation pressure shall be between 2,700 and 3,000 psig.

4.6.13 Oxygen bomb test.

4.6.13.1 Test setup. The test setup used in performing the bomb tests shall be capable of delivering a minimum pressure of 2,300 psig. When opened, the test setup shall deliver at least 95% of the total pressure rise in 20 $^{+}_{-}$7 milliseconds at the valve being tested. The pressure shall be delivered through a 0.188 inch ID steel tube approximately 6 inches long (see 6.10). The test shall contain an off-line device capable of releasing pressure without removing the valve.

4.6.13.2 Procedure. The valve inlet (0.5 or 1.0 inch ANPT) shall be sealed against a minimum pressure of 2,300 psig and the outlet shall be installed on the test setup. With the valve in the open position, the test setup shall be activated (opened) and the valve pressurized to a minimum pressure of 2,300 psig. This pressure shall be maintained for 30 seconds and then released. This operation shall be repeated 3 times. There shall be no indication of internal or external charring or deterioration. The valve shall then be subjected to the torque, flow, and leakage tests (see 4.6.4, 4.6.5, and 4.6.6).

4.6.14 Salt fog. When specified (see 6.2), the valve shall be subjected to the salt fog test in ASTM-B117 for 50 hours. There shall be no visible damage to the valve. The valve shall then be subjected to the torque and leakage test (see 4.6.4 and 4.6.6).

4.6.15 Cap. The cap shall be subjected to a 100 lb pull. The cap shall remain attached to the tether and the tether shall remain attached to the valve.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The valves covered by this specification are intended for use in high-pressure, fixed and portable, aeronautical oxygen systems for filling and controlling the flow from these systems.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Item identification and valve size.
c. Packaging requirements (see 5.1).
d. If the conformance inspection sampling inspection level is other than normal (tightened or reduced) (see 4.3).
e. The requirement for the vendor to identify proposed cleaning and verification methods (see 4.6.3).
f. Whether the salt fog test is waived and material certification is acceptable (see 4.6.14).
g. When first article is required (see 3.1).

6.3 Cleaning. See MIL-STD-1330 and SAE-ARP1176 for guidance on proven cleaning methods and verification. Visible inspections are typically conducted using white light and ultraviolet light to detect particulates and some types of hydrocarbon. The non-volatile residue test in SAE-ARP1176 may be used to baseline the hydrocarbon verification at an acceptable contamination level, however, other cleaning and verification methods that do not contain class I and II ODC solvents should be used for production.

6.4 Flow test. A proven procedure to maintain inlet and outlet pressure as specified is to use a pressurized surge tank at both the inlet and outlet to stabilize flow.

6.5 Lubrication. MIL-PRF-27617 (type III) is a lubricant proven compatible with pure oxygen.

6.6 SLPM. One SLPM is equal to one liter of flow per minute at a pressure of 760 millimeters of mercury at the temperature of 0°Celsius.

6.7 Siphon tube. A previously acceptable design of this valve had a siphon tube, with radially distributed holes, installed to prevent any foreign matter within the cylinder from being carried into the valve.

6.8 Internal isolation device. A previously acceptable design of this valve used a metal diaphragm to separate the threaded handwheel shaft from the internal moving components. The

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metal diaphragm kept the lubrication and contaminant on the handwheel shaft from entering the valve and creating a fire hazard.

6.9 Breakaway of bubbles. The breakaway of bubbles is the formation and breaking away of bubbles on the surface of a submerged test valve while applying internal pressure. The formation of bubbles is acceptable but breaking away is not.

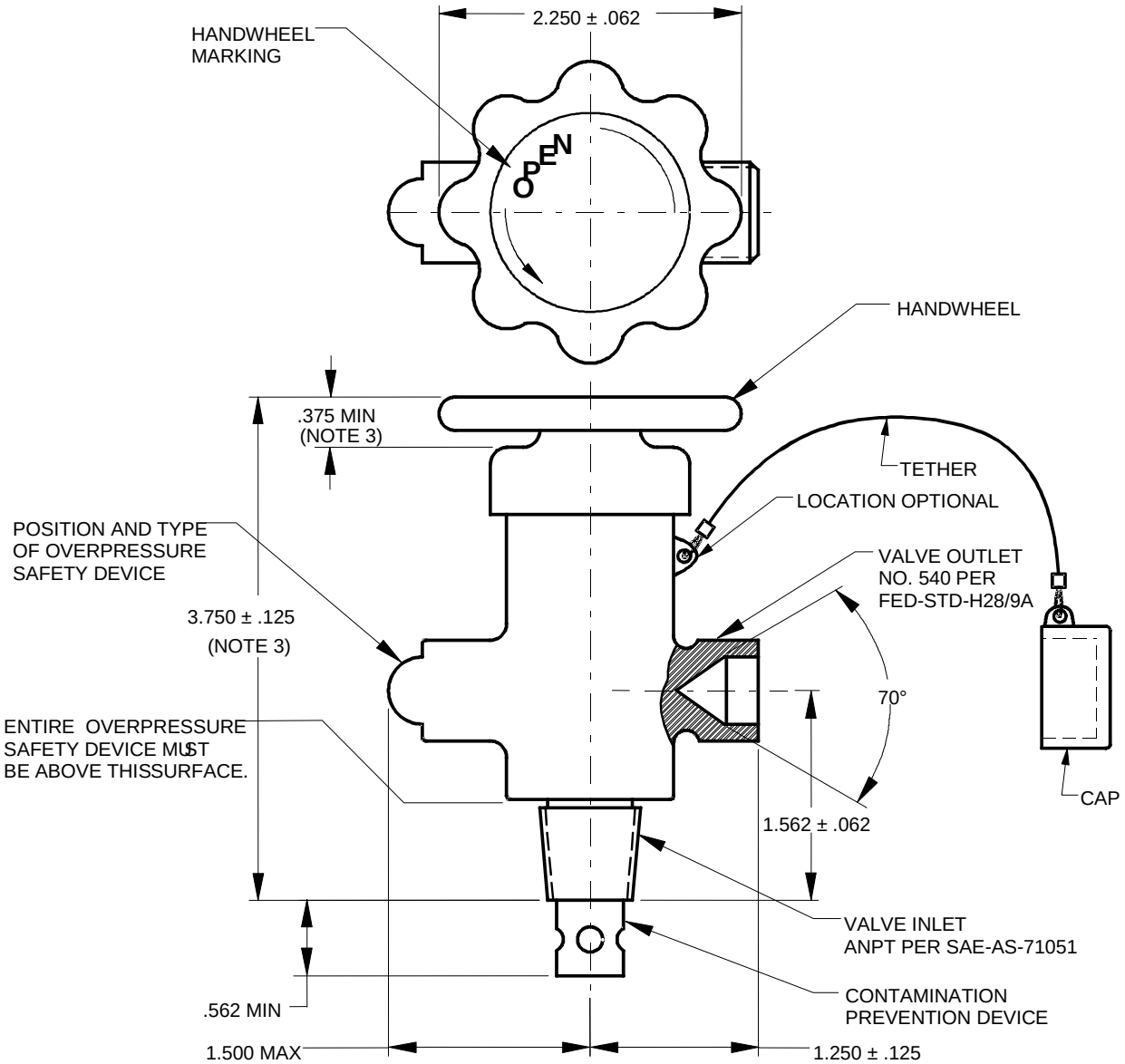
6.10 Oxygen bomb test. A previously acceptable design of this pressure delivery system was a 0.188 inch ID, corrosion-resistant steel tube approximately 6 inches long with a bend of less than 4 degrees.

6.11 Subject term (key word) listing.

Breathing
Shut-off
2,100 psig system

6.12 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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PIN	INLET PIPE THREAD	MAX WEIGHT
M9439-1	SIZE 1 .5 NPT	20 OZ.
M9439-2	SIZE 2 1 NPT	26 OZ.

NOTES: 1. DIMENSIONS IN INCHES, UNLESS OTHERWISE SPECIFIED.

2. TOLERANCES: UNLESS OTHERWISE SPECIFIED: DECIMALS ± .005
ANGLES ± .5°
3. VALVE SHOWN IN CLOSED POSITION.

4. FIGURE NOT TO SCALE.

FIGURE 1. Oxygen cylinder valve

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Air Force - 71
Army - AV
Navy - AS

Preparing Activity:
Air Force - 71

(Project 4820-2009-022)

Review Activities:
Air Force - 11, 99
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